



DEALER BEST PRACTICE SERIES

Improving CRC Rebuild Turnaround Time

Application Maintenance Site Management **Component Rebuild** Safety MARC Management

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Improving CRC Rebuild Turnaround Time	DATE 6/9/2017	CHG NO 01	NUMBER 0906-4.5-1013

1.0 Introduction

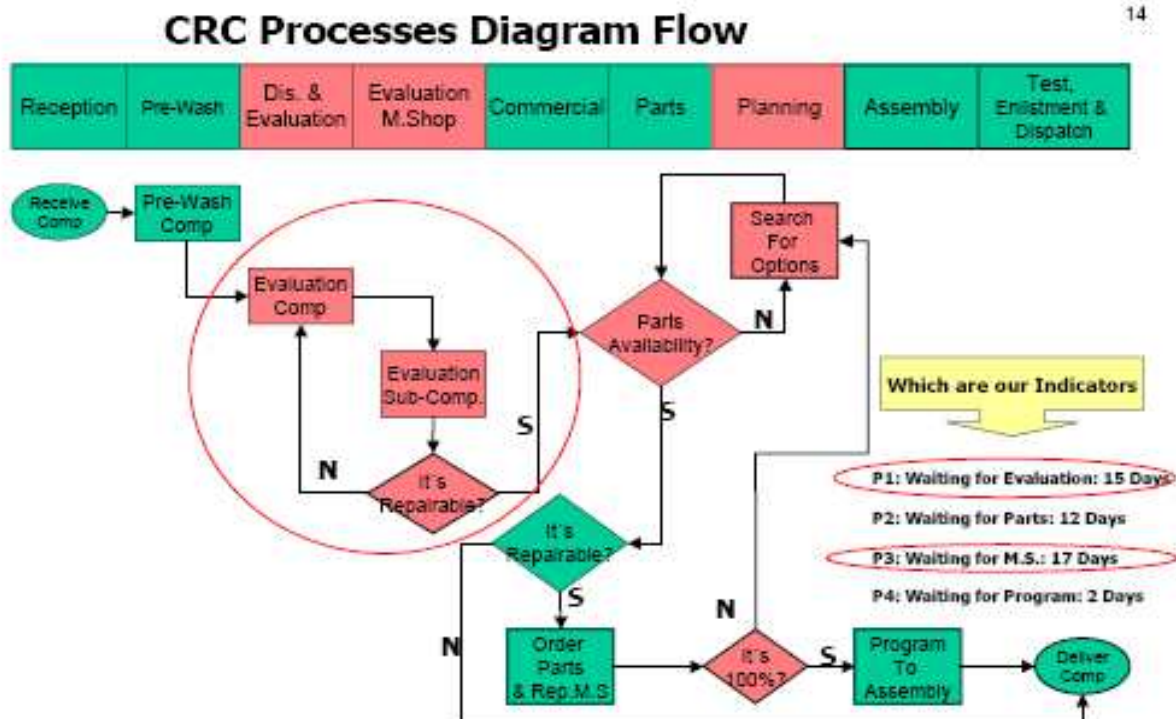
The time required by a Component Rebuild Center to complete a major component rebuild in support of Mining fleets is a key performance metric that has substantial impact on CRC financials. The turnaround time is defined as the time a component enters a shop for rebuild to the time it is ready to ship. Managing and reducing turnaround time requires high efficiency, proper tooling and equipment, solid processes, and smooth coordination and planning. Since there are many variables to consider and possibly many root causes of delays to consider, efforts to improve turnaround time are best addressed as a 6Sigma project. This Best Practice contains information and key findings gathered from several dealers who have used 6Sigma to reduce CRC component turnaround time.

Benchmarks for a turnaround time target goals contained in [\(archived\) CRC Develop Guide \(SEBF8813\)](#), was 20 days for 3500 series engines, and 10 days for all other components. These goals can vary depending on regional factors such as time to order and receive parts, machine shop capability and cycle time, number of shifts, and other factors.

2.0 Best Practice Description

Improving Component Turnaround Time makes an excellent 6Sigma project (DMAIC).

The first step in improving CRC component turnaround time is to define the "as is" baseline. This means mapping the current process from the time a component enters the CRC to the time the component is ready to ship. Time spent in each major step is documented.



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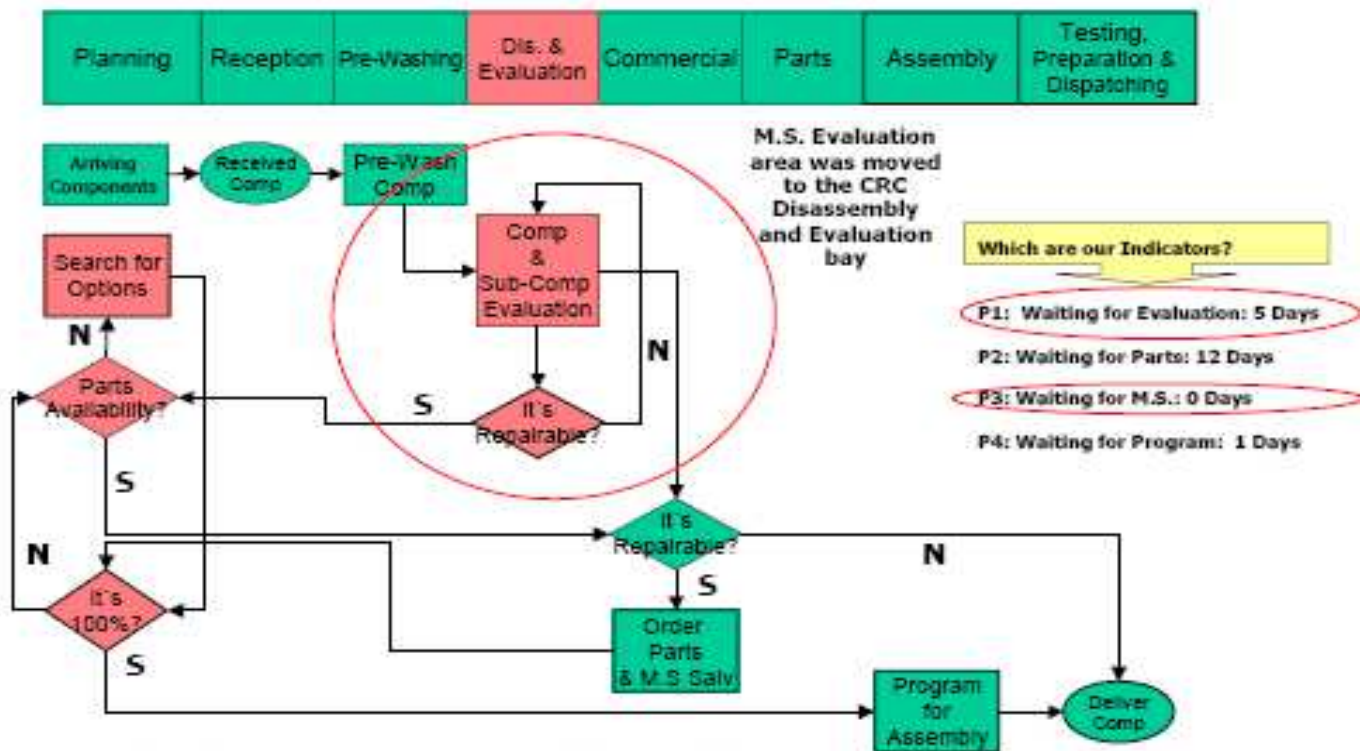
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(Process-Mapping contd.)

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CRC Processes Diagram Flow



Typically, the component processes documented and measured include engines, transmissions, wheel stations, torque converters, and differentials. Other components may be also included and engine turnaround time is often measured by engine family.

The major steps in the process include:

- Planning & Scheduling
- Build Kits
- Receiving
- Pre-Wash
- Disassemble
- Evaluate
- Quote (if required)
- Machine Shop Evaluation and Salvage
- Assemble
- Test
- Dispatch

Once the process is mapped and time spent in each step is documented, the **Define** and **Measure** steps of the 6Sigma process are essentially complete.

The next step is **Analyze**.

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The purpose of the **Analyze** phase is to identify delays and interruptions in the rebuild process. **Root causes of the delays are then identified and solution options are evaluated.**

Here are examples of some of the delays and solutions dealers have identified.

- Waiting for the machine shop to complete salvage work can take 10 or more days for major parts such as engine blocks, wheel station spindles, crankshafts, etc.
 - Solution 1: The machine shop purchased an inventory of exchange piece parts. If a key part, such as an engine block, took more than 5 days to salvage and turned more than 10 times a year, an "exchange" part was stocked.
 - Solution 2: A machine shop parts inspector/evaluator was assigned to the CRC disassembly and evaluation area to inspect and assess any needed machine shop work as the component was being disassembled. He would schedule and coordinate the work.
- The "quotation" process and getting customer work order approval can cause substantial delays.
 - Solution 1: As much as possible, develop flat rates with pre-approved work orders.
 - Solution 2: Establish processes with the customer to accelerate the approval processes when required to a limit of no more than 3 days.
- Common "waiting" time lost.
- Waiting for information.
 - Solution: Install wireless SIS terminals in the shop. One terminal per every 2 adjacent bays minimum.
- Waiting for parts.
 - Solution: Establish must do parts kits for common jobs. For planned component rebuilds, schedule and pre-open a work order so that must do parts are pre-positioned in the shop. Parts are delivered directly to the work stations. Mechanics do not have to run for parts.
- Looking for parts or tools.
 - Solution: Organized parts storage racks / carts with assigned spots. Required special tools permanently at the work station.

Other common issues and solutions include more efficient tooling, such as cleaning equipment or component rebuild/ rollover stands and improved material handling.

Improving and managing component turnaround time requires high standards of efficiency and well-designed processes. It has substantial impact on the financial performance of a service operation and the optimum utilization of assets. See attached PowerPoint presentation of dealer 6Sigma project summary.

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3.0 Implementation Steps

Due to the potential of multiple root causes for component rebuild process bottlenecks, the issue makes a good 6Sigma project (DMAIC). The key steps are:

- Define: Create a CRC process map.
- Measure: Determines the days spent in each major step of the rebuild process.
- Analyze: Identify unnecessary bottlenecks and causes for delays.
- Improve: Change / modify processes
- Control: Deploy and measure improvement.

4.0 Benefits

Major component rebuild turnaround time has substantial impact on a service operation financial performance. The experiences of several dealers who have successfully improved turnaround time show lowered costs, better use of assets, better quality, improved capacity, lower exchange component inventory requirements, and improved cash flow. The following are examples of reported benefits:

- Average labor hours spent per component reduced 20% - 35%, thus reducing expenses.
- By reducing turnaround time, one dealer cancelled plans for a \$1.5M shop expansion. Capacity improvement was able to handle workloads without the need to add shop floor space and equipment.
- Another dealer was able to reduce the size of their exchange component inventory supporting 793 fleets by 20% due to improved shop turnaround.
- Time from first work to invoice improved along with turnaround time, creating a cash flow improvement.
- Several dealers report lower re-do costs when rebuilds flow smoothly without interruptions and delays.

5.0 Resources Required

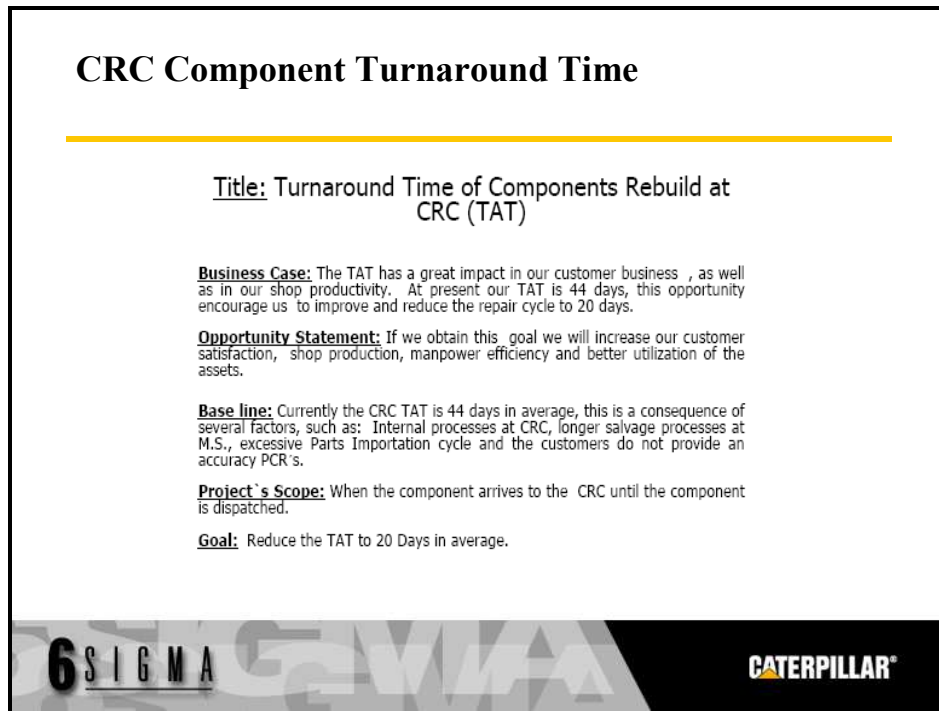
Improvements to CRC component turnaround time may be impacted by any facet of service operations including tooling, facility size, layout, material salvage, part inventories, technician availability, etc. Solutions to improve turn around times, which may require an investment should best be considered on a cost versus benefit basis.

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6.0 Supporting Attachments

Refer to [Component Turnaround Time](#), MS- Powerpoint presentation included in the “Attachments” section of this file. (Click on “Attachments” in this pdf file)

- Slides are from an actual 6Sigma project, launched to improve CRC Component Turnaround Time.



Component Turnaround Time.ppt

7.0 Related Best Practices

Utilizing Exchange Program for Slow-Turn Salvaged Parts to Improve CRC TAT 1006-4.2-1030
 Labor Efficiency Lowers CRC Costs 0207-4.5-1062
 Using Cameras to Conduct CRC Process Reviews 0408-4.1-1117

8.0 Acknowledgements

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